Lab_4_VibeCoding...
PDF

https://github.com/SETAPESU26/19_battleship

where do i tdo it

>> git clone https://github.com/SETAPESU26/19_battleship.git
git : The term 'git' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:2 char:1
+ git clone https://github.com/SETAPESU26/19_battleship.git
+ ~~~
    + CategoryInfo          : ObjectNotFound: (git:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

(.venv) PS C:\Users\User\Downloads>

which one

PS C:\Users\User\Downloads\sastrugi> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\Users\User\Downloads\sastrugi\.venv\Scripts\Activate.ps1)

(.venv) PS C:\Users\User\Downloads\sastrugi> cd $HOME\Downloads
>> git clone https://github.com/SETAPESU26/19_battleship.git
Cloning into '19_battleship'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 11 (delta 0), reused 0 (delta 0), pack-reused 8 (from 1)
Receiving objects: 100% (11/11), done.
Resolving deltas: 100% (2/2), done.
(.venv) PS C:\Users\User\Downloads>

Scenario 19 — Battleship vs AI

A terminal Battleship game with separate board and AI modules.

Provided files

- `main.py` — entry point.
- `game.py` — turn flow and setup.
- `board.py` — ship and shot state.
- `ai.py` — computer targeting.
- `requirements.txt` — dependency declaration.

Setup

```bash
python main.py
```

Before changing the code

Inspect how coordinates are represented in every module. Fire at known ship cells and
known empty cells. Then trace an AI shot from selection to hit detection.

Task 1 — Consistent coordinate handling

Make coordinate representation consistent across input, ship placement, player shots,
and AI shots so occupied cells are recognised correctly.

Done when: every real ship coordinate is treated as a hit and every known empty
coordinate is treated as a miss.

Task 2 — Complete fleet and win logic

Support multiple ships, track individual ship damage, identify sunk ships, prevent
repeated shots, and end the game when the fleet is sunk.

Task 3 — Improve the AI

After a hit, make the AI preferentially consider nearby untried cells. It must never
shoot the same coordinate twice and must handle a board with no remaining choices.

Task 4 — Shot-level feedback

Ensure hit/miss/sunk feedback occurs exactly once for an actual shot. Internal AI
candidate selection must not produce false hit/miss messages.

Required testing

Test hits, misses, repeated shots, sinking one ship, sinking all ships, AI repeated-shot
prevention, adjacent targeting after a hit, invalid coordinates, and quitting.

LLM usage

You may use an LLM during the lab. The goal is to use it as a coding assistant while
retaining responsibility for understanding and testing the result.

- Inspect the existing code before asking for changes.
- Ask for explanations when you do not understand a proposed change.
- Test generated code against the stated behaviour and edge cases.
- Keep your complete LLM chat history for submission.
- Do not replace the whole project with an unrelated implementation.
- Keep all state in memory; do not add CSV, JSON, SQLite, or other persistence.

Submission checklist

- [ ] Task 1 completed and the original defect was reproduced and fixed.
- [ ] Tasks 2–4 completed and tested.
- [ ] Boundary and invalid-input cases tested.
- [ ] No unnecessary external dependencies added.
- [ ] No persistent storage added.
- [ ] Code remains understandable and modular.
- [ ] Complete LLM chat-history link included.

Folder structure

```text
scenario-07-battleship/
├── README.md
├── requirements.txt
├── main.py
├── game.py
├── board.py
```

└── ai.py
```

## Submission Checklist

Submission is only the following three things:

- [ ] A 10-second video of gameplay **before** your changes, showing the bug/broken behavior
- [ ] A 10-second video of gameplay **after** your changes, showing the bug fixed and the new features working
- [ ] The Chat/LLM used page link, with the complete chat history

Worked for 8s

```
PS C:\Users\User\Downloads\19_battleship> cd
$HOME\Downloads\19_battleship
>> dir

    Directory: C:\Users\User\Downloads\19_battleship

Mode   LastWriteTime   Length Name
----   -------------   ------ ----
-a----   10/11/2026   6:55 PM   421 ai.py
-a----   10/11/2026   6:55 PM   340 board.py
-a----   10/11/2026   6:55 PM   1805 game.py
-a----   10/11/2026   6:55 PM   83 main.py
-a----   10/11/2026   6:55 PM   3137 README.md
-a----   10/11/2026   6:55 PM   112 requirements.txt

PS C:\Users\User\Downloads\19_battleship>

PS C:\Users\User\Downloads\19_battleship> python main.py
Python was not found; run without arguments to install from the
Microsoft Store, or disable this shortcut from Settings > Apps >
```
```

Advanced app settings > App execution aliases.
PS C:\Users\User\Downloads\19_battleship>

PS C:\Users\User\Downloads\19_battleship> py --version
Python 3.12.7
PS C:\Users\User\Downloads\19_battleship>

Your shots are coordinates like 2,3.
Ship cells remaining: 3
>

```
----            -------------       ------ ----
-a----        10/11/2026   6:55 PM            421 ai.py
-a----        10/11/2026   6:55 PM            340 board.py
-a----        10/11/2026   6:55 PM           1805 game.py
-a----        10/11/2026   6:55 PM             83 main.py
-a----        10/11/2026   6:55 PM           3137 README.md
-a----        10/11/2026   6:55 PM            112 requirements.txt
```

PS C:\Users\User\Downloads\19_battleship> python main.py
Python was not found; run without arguments to install from the Microsoft Store, or disable this shortcut from Settings > Apps > Advanced app settings > App execution aliases.
PS C:\Users\User\Downloads\19_battleship> py --version
Python 3.12.7
PS C:\Users\User\Downloads\19_battleship> py main.py
Battleship

Your shots are coordinates like 2,3.
Ship cells remaining: 3
> 2,3
MISS!
AI fired at 3,6

Your shots are coordinates like 2,3.

Ship cells remaining: 3
> 0,0
Outside board.

Your shots are coordinates like 2,3.
Ship cells remaining: 3
> 4,5
MISS!
AI fired at 6,6

Your shots are coordinates like 2,3.
Ship cells remaining: 3
>
i recorded it too 20 sec video

Worked for 5s

```
class Board:
    SIZE = 6

    def __init__(self):
        self.ships = set()
        self.shots = set()

    def place_ship(self, cells):
        self.ships.update(cells)

    def fire(self, pos):
        self.shots.add(pos)
        return pos in self.ships

    def all_sunk(self):
        return self.ships <= self.shots
```

```
from board import Board
from ai import AI
```

```
class Battleship:
    def __init__(self):
        self.player = Board()
        self.enemy = Board()
        self.ai = AI()
        self._setup()

    def _setup(self):
        self.player.place_ship({(1, 1), (1, 2), (1, 3)})
        self.enemy.place_ship({(2, 2), (2, 3), (2, 4)})

    def show(self):
        print("\nYour shots are coordinates like 2,3.")
        print("Ship cells remaining:", len(self.enemy.ships -
self.player.shots))

    def run(self):
        print("Battleship")
        while True:
            self.show()
            raw = input("> ").strip().lower()
            if raw == "q":
                return
            try:
                r, c = map(int, raw.split(","))
                pos = (r - 1, c - 1)
            except ValueError:
                print("Use row,col.")
                continue
            if not (0 <= pos[0] < Board.SIZE and 0 <= pos[1] <
Board.SIZE):
                print("Outside board.")
                continue
            if pos in self.player.shots:
                print("Already fired there.")
                continue
            print("HIT!" if self.enemy.fire(pos) else "MISS!")
```

```
            if self.enemy.all_sunk():
                print("You sank the fleet.")
                return

            ai_pos = self.ai.choose()

            # representation consistent through the whole flow.
            try:
                ar, ac = map(int, ai_pos.split(","))
                player_pos = (ar, ac)
            except ValueError:
                player_pos = None
            if player_pos is not None:
                print("AI fired at", ai_pos)
                if player_pos in self.player.ships:
                    print("AI scored a hit.")

import random


class AI:
    def __init__(self, size=6):
        self.size = size
        self.tried = set()

    def choose(self):
        options = [(r, c) for r in range(self.size) for c in range(self.size)
                   if (r, c) not in self.tried]
        pos = random.choice(options)
        self.tried.add(pos)
        return f"{pos[0] + 1},{pos[1] + 1}"  # intentional representation
mismatch
```

instead of using copilot cant u jus give me the codes

i jus changed all the files now i should do py main.py right and should i record again or

PS C:\Users\User\Downloads\19_battleship> python main.py
Python was not found; run without arguments to install from the Microsoft Store, or disable this shortcut from Settings > Apps > Advanced app settings > App execution aliases.
PS C:\Users\User\Downloads\19_battleship> py --version
Python 3.12.7
PS C:\Users\User\Downloads\19_battleship> py main.py
Battleship

Your shots are coordinates like 2,3.
Ship cells remaining: 3
> 2,3
MISS!
AI fired at 3,6

Your shots are coordinates like 2,3.
Ship cells remaining: 3
> 0,0
Outside board.

Your shots are coordinates like 2,3.
Ship cells remaining: 3
> 4,5
MISS!
AI fired at 6,6

Your shots are coordinates like 2,3.
Ship cells remaining: 3
> Traceback (most recent call last):
  File "C:\Users\User\Downloads\19_battleship\main.py", line 4, in <module>

  File "C:\Users\User\Downloads\19_battleship\game.py", line 24, in run

```
    def show(self):
                ^^
KeyboardInterrupt
PS C:\Users\User\Downloads\19_battleship> py main.py
Battleship vs AI

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 7
Your ship cells remaining: 6
> 3,3
You HIT!
AI fired at 6,6
AI SUNK a ship!

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 6
Your ship cells remaining: 5
> 3,4
You HIT!
AI fired at 5,6
AI MISS!

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 5
Your ship cells remaining: 5
> 3,5
You SUNK a ship!
AI fired at 6,5
AI MISS!

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 4
Your ship cells remaining: 5
```

> 3,3
Already fired there.

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 4
Your ship cells remaining: 5
> 0,0
Outside board. Use coordinates from 1 to 6.

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 4
Your ship cells remaining: 5
> abc
Invalid input. Enter row,col (e.g. 2,3) or q.

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 4
Your ship cells remaining: 5
> q
Game quit.
PS C:\Users\User\Downloads\19_battleship>

so record a video now? what should i enter in the chat

Screen Recording ...
File

after

Worked for 24s

ok git

https://github.com/mano-glitch-ai/19_battleship

```
> q
Game quit.
PS C:\Users\User\Downloads\19_battleship> cd
$HOME\Downloads\19_battleship
PS C:\Users\User\Downloads\19_battleship> git remote -v
origin  https://github.com/SETAPESU26/19_battleship.git (fetch)
origin  https://github.com/SETAPESU26/19_battleship.git (push)
PS C:\Users\User\Downloads\19_battleship> git remote rename
origin upstream
PS C:\Users\User\Downloads\19_battleship> git remote add origin
https://github.com/mano-glitch-ai/19_battleship.git
PS C:\Users\User\Downloads\19_battleship> git remote -v
origin  https://github.com/mano-glitch-ai/19_battleship.git (fetch)
origin  https://github.com/mano-glitch-ai/19_battleship.git (push)
upstream        https://github.com/SETAPESU26/19_battleship.git
(fetch)
upstream        https://github.com/SETAPESU26/19_battleship.git
(push)
PS C:\Users\User\Downloads\19_battleship> git add .
>> git status
>> git commit -m "Fix Battleship gameplay and AI"
>> git push -u origin HEAD
On branch main
Your branch is up to date with 'upstream/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
        new file:   __pycache__/ai.cpython-312.pyc
        new file:   __pycache__/board.cpython-312.pyc
        new file:   __pycache__/game.cpython-312.pyc
        modified:   ai.py
        modified:   board.py
        modified:   game.py
        modified:   main.py

Author identity unknown
```

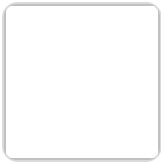
*** Please tell me who you are.

Run

```
git config --global user.email "you@example.com"
git config --global user.name "Your Name"
```

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'User@Manoj.(none)')
info: please complete authentication in your browser...
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 32 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (11/11), 3.88 KiB | 3.88 MiB/s, done.
Total 11 (delta 2), reused 11 (delta 2), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/mano-glitch-ai/19_battleship.git
 * [new branch]   HEAD -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\User\Downloads\19_battleship>



itgs already here why should i change all that

Enemy ship cells remaining: 4
Your ship cells remaining: 6
> 0,0
Outside board. Use coordinates from 1 to 6.

Your shots are coordinates like 2,3.

```
Enter q to quit.
Enemy ship cells remaining: 4
Your ship cells remaining: 6
> q
Game quit.
PS C:\Users\User\Downloads\19_battleship> cd
$HOME\Downloads\19_battleship
PS C:\Users\User\Downloads\19_battleship> git remote -v
origin  https://github.com/SETAPESU26/19_battleship.git (fetch)
origin  https://github.com/SETAPESU26/19_battleship.git (push)
PS C:\Users\User\Downloads\19_battleship> git remote rename
origin upstream
PS C:\Users\User\Downloads\19_battleship> git remote add origin
https://github.com/mano-glitch-ai/19_battleship.git
PS C:\Users\User\Downloads\19_battleship> git remote -v
origin  https://github.com/mano-glitch-ai/19_battleship.git (fetch)
origin  https://github.com/mano-glitch-ai/19_battleship.git (push)
upstream        https://github.com/SETAPESU26/19_battleship.git
(fetch)
upstream        https://github.com/SETAPESU26/19_battleship.git
(push)
PS C:\Users\User\Downloads\19_battleship> git add .
>> git status
>> git commit -m "Fix Battleship gameplay and AI"
>> git push -u origin HEAD
On branch main
Your branch is up to date with 'upstream/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
        new file:   __pycache__/ai.cpython-312.pyc
        new file:   __pycache__/board.cpython-312.pyc
        new file:   __pycache__/game.cpython-312.pyc
        modified:   ai.py
        modified:   board.py
        modified:   game.py
```

```
        modified:   main.py

Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

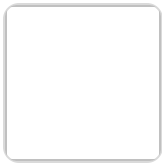
to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'User@Manoj.(none)')
info: please complete authentication in your browser...
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 32 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (11/11), 3.88 KiB | 3.88 MiB/s, done.
Total 11 (delta 2), reused 11 (delta 2), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/mano-glitch-ai/19_battleship.git
 * [new branch]      HEAD -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\User\Downloads\19_battleship> git config user.name
"Manoj Kumar S"
>> git config user.email "manosen2006@gmail.com"
PS C:\Users\User\Downloads\19_battleship> git add README.md
ai.py board.py game.py main.py
>> git commit -m "Fix Battleship gameplay and AI"
>> git push origin main
[main af324f0] Fix Battleship gameplay and AI
 8 files changed, 160 insertions(+), 28 deletions(-)
 create mode 100644 __pycache__/ai.cpython-312.pyc
 create mode 100644 __pycache__/board.cpython-312.pyc
```

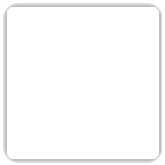
```
 create mode 100644 __pycache__/game.cpython-312.pyc
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 32 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (11/11), 7.61 KiB | 3.80 MiB/s, done.
Total 11 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mano-glitch-ai/19_battleship.git
   68e3f33..af324f0  main -> main
PS C:\Users\User\Downloads\19_battleship>
```

ok after this

where should i put each of them tell me step by step



now what



yeh what to do

i already did that before

startefd

```
Total 11 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mano-glitch-ai/19_battleship.git
   68e3f33..af324f0  main -> main
```

```
PS C:\Users\User\Downloads\19_battleship> "__pycache__/" | Set-
Content .gitignore
>> "*.py[cod]" | Add-Content .gitignore
>> git rm -r --cached __pycache__
>> git add .gitignore
>> git commit -m "Remove Python cache files"
>> git push origin main
rm '__pycache__/ai.cpython-312.pyc'
rm '__pycache__/board.cpython-312.pyc'
rm '__pycache__/game.cpython-312.pyc'
[main b63bb17] Remove Python cache files
 4 files changed, 2 insertions(+)
 create mode 100644 .gitignore
 delete mode 100644 __pycache__/ai.cpython-312.pyc
 delete mode 100644 __pycache__/board.cpython-312.pyc
 delete mode 100644 __pycache__/game.cpython-312.pyc
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 32 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 315 bytes | 315.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mano-glitch-ai/19_battleship.git
   af324f0..b63bb17  main -> main
PS C:\Users\User\Downloads\19_battleship> py main.py
Battleship vs AI

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 7
Your ship cells remaining: 6
> 3,3
You HIT!
AI fired at 1,2
AI MISS!
```

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 6
Your ship cells remaining: 6
> 3,4
You HIT!
AI fired at 2,1
AI MISS!

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 5
Your ship cells remaining: 6
> 3,5
You SUNK a ship!
AI fired at 1,6
AI MISS!

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 4
Your ship cells remaining: 6
> 1,5
You HIT!
AI fired at 6,6
AI SUNK a ship!

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 3
Your ship cells remaining: 5
> 1,6
You SUNK a ship!
AI fired at 6,5
AI MISS!

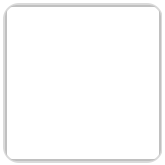
Your shots are coordinates like 2,3.

```
Enter q to quit.
Enemy ship cells remaining: 2
Your ship cells remaining: 5
> 5,1
You HIT!
AI fired at 5,6
AI MISS!

Your shots are coordinates like 2,3.
Enter q to quit.
Enemy ship cells remaining: 1
Your ship cells remaining: 5
> 5,2
You SUNK a ship!
You sank the entire enemy fleet. You win!
PS C:\Users\User\Downloads\19_battleship>
```

what do to do now

```
PS C:\Users\User\Downloads\19_battleship> py -c "from ai import
AI; a=AI(6); a.tried={(r,c) for r in range(6) for c in range(6) if (r,c)!=
(2,2) and (r,c)!=(2,3)}; a.record_result((2,2),'hit'); print('Next AI shot:',
a.choose(), '(expected adjacent cell (2,3))')"
Next AI shot: (2, 3) (expected adjacent cell (2,3))
PS C:\Users\User\Downloads\19_battleship>
```



i did the chat history thing see

done

Lab 4: VibeCoding
Objective

Use Vibe Coding tools to fix the broken code and add features to the same. Learn how
to prompt to get the job done in under three to four attempts. Clone a repo assigned
to you, make changes and push to your repo in your personal account.
THIS IS AN INDIVIDUAL ASSIGNMENT. WORK INDIVIDUALLY!!
Duration
60 minutes
Software Requirements
● Access to Vibe Coding tool. (ChatGPT, Gemini, Claude etc)
Instructions:
1. Go to the repo assigned to you
2. Go through the ReadMe file to understand the deliverables
3. Clone or Fork the repo
4. Run the Python based code (A game/ A Puzzle)
5. Record 10 seconds of video before making any changes.
6. Write prompts and fix the broken code
7. Write prompts to add all the features listed in the ReadMe file of the repo one
by one.
8. Record another 10 seconds of video again after making all the changes.
9. Push the code to your repo.
10.Do not raise PR to the main repo (SETAPESU26)
Deliverables:
Upload the following to your Lab repo under Lab-4 folder,
a. Videos (before and after)
b. Updated Code
c. Chat history exported as a doc/pdf

how to make it a doc or pdf

See another version